MILITARY SPECIFICATION

VEHICLE, COMBAT ENGINEER FULL TRACKED:
165MM, GUN, M728

This amendment forms a part of Military Specification MIL-V-62008B(AT), dated 16 May 1983, and is approved for use by the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1, under "STANDARDS", "MILITARY" delete:

"MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-210	- Climatic Extremes for Military Equipment."

2.1.2, under "DRAWINGS", "ARMY" delete:

"10940104	- Installation, Mount, Combination Gun, M150."

PAGE 4

3.4, delete second sentence and substitute:

"The vehicle and all parts requiring identification shall be marked in accordance with MIL-STD-130 and MIL-STD-642."

PAGE 5

3.4.4.4, add new paragraph to document as follows:

"3.4.4.4 Fuel shutoff valve. The engine shall stop firing and rotating in not more than 30 seconds after the manual fuel shut-off valve is actuated to the "off" position."

PAGE 24

4.2.3, delete 4.2.3 and substitute: "4.2.2.3"

4.2.3.1, delete 4.2.3.1 and substitute: "4.2.2.3.1"

FSC 2350

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4.3, add the following subparagraph:

"k. Tests in table VIII annotated * (see 4.5.2)."

4.5, subparagraph a, delete 4.5.1.43.5 and substitute: "4.5.1.47"

4.5.1, delete in its entirety, and substitute:

"4.5.1 Acceptance inspections."

4.5.1.1, delete the first two sentences and substitute:

"4.5.1.1 Acceptance tests. To determine conformance to 3.6 after break-in run (see 3.5), each vehicle shall be examined for evidence that in-process inspection of 4.3 has been performed. The vehicle shall then be operated from a distance of not less than 8 miles and subjected to acceptance tests specified at place of manufacture in table VIII.

PAGES 26, 27, AND 28

Table VIII, delete in its entirety and substitute the following:

"Table VIII. Location for tests.

Title	Paragraph No.	Place of manufacture	Government proving ground
Acceptance tests	4.5.1.1	X	X
Sealing check	4.5.1.2.1	*	X
Hatch seal test	4.5.1.3	*	X
Fuel system slope check	4.5.1.4	X	X
Fuel line check	4.5.1.5.1	X	X
Hydraulic system check	4.5.1.5.2	X	X
Heater fuel feed check	4.5.1.6	X	X
Control checks	4.5.1.7	X	X
Adjustment mechanism check	4.5.1.8	X	X
Drivers hatch and escape hatch check	4.5.1.9	*	X
Loaders and commanders hatch check	4.5.1.10	*	X
Seating check	4.5.1.12	*	X
Light check	4.5.1.13	X	X
Passive driver's night viewer power circuit check	4.5.1.16	*	X
Slip ring and external communication circuit check	4.5.1.17	X	X

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Table VIII. Location for tests – Continued.

Title	Paragraph No.	Place of manufacture	Government proving ground
Auxiliary outlet check	4.5.1.18	*	X
Searchlight control system check	4.5.1.19	*	X
Power plant electrical check	4.5.1.20	*	X
Air cleaner blower check	4.5.1.21	X	X
Engine manifold heater check	4.5.1.22	X	X
Personnel heater	4.5.1.23	X	X
Engine smoke generator test	4.5.1.24	*	X
Communication system check	4.5.1.25	*	X
Main gun firing circuits check	4.5.1.26	X	X
Machine gun firing circuits check	4.5.1.27	X	X
Main gun and machine gun circuit control check	4.5.1.28	X	X
Smoke grenade discharger circuit check	4.5.1.29 thru 4.5.1.29.3	*	X
Power package and train test	4.5.1.30	X	X
Level road speed test	4.5.1.31	X	X
Acceleration test	4.5.1.32	X	X
Engine starting on grades and slopes test	4.5.1.33	X	X
Stopping test	4.5.1.34	X	X
Holding test	4.5.1.35	X	X
Turning test	4.5.1.36	X	X
Fording test	4.5.1.37	X	X
Drain valve checks	4.5.1.38	*	X
Inflatable seal and pump check	4.5.1.39	*	X
Turret and gun control system check	4.5.1.40	X	X
Main armament balance	4.5.1.40.1	*	
Nylon ballistic shield check	4.5.1.40.2	*	X
Manual traverse effort test	4.5.1.40.3	X	X
Manual elevation effort test	4.5.1.40.4	X	X
Manual elevation and depression response rate test	4.5.1.40.5	X	X
System backlash test	4.5.1.40.6	*	X
Simultaneous use of power and manual control test	4.5.1.40.7	X	X
Override control test	4.5.1.40.8	X	X
Control system deadspot test	4.5.1.40.9	X	X
Gun elevation speed test	4.5.1.40.10	X	X
Turret traversing speed test	4.5.1.40.11	X	X
Main gun stability test	4.5.1.40.12	*	X

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TABLE VIII. Location for tests – Continued.

Title	Paragraph No.	Place of manufacture	Government proving ground
Elevation and depression limit tests	4.5.1.40.13	X	X
Gun laying on stationary target test	4.5.1.40.14	*	X
Operation on slope test	4.5.1.40.15	X	X
Turret locking test	4.5.1.40.16	X	X
Azimuth indicator test	4.5.1.40.17	X	X
Turret control system fluid leaking test	4.5.1.40.18	X	X
Hydraulic pressure limit test	4.5.1.40.19	X	X
Gun fire control system test	4.5.1.41	*	
Synchronization test	4.5.1.41.1 thru 4.5.1.41.1.2	*	X
		*	X
Sighting system backlash	4.5.1.41.2	*	X
Boresight knob travel test	4.5.1.41.3	*	X
Gunner/Commander night vision	4.5.1.42.7	*	X
Boom check	4.5.1.43.1	X	X
Winch check	4.5.1.43.2	X	X
Winch brake test	4.5.1.43.3	X	X
Bulldozer test	4.5.1.43.4	X	X
Vehicle hydraulic system fluid leakage check	4.5.1.43.5	X	X
Emergency mode firing	4.5.1.44	X	X
Elevation shutoff valve	4.5.1.45	X	X
Turret traverse brake	4.5.1.46	X	X
Fuel shutoff valve	4.5.1.47	X	X
Control tests	4.5.2	X	X
Contamination check	4.5.2.3	X	X

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TABLE VIII. Location for tests – Continued.

Title	Paragraph No.	Place of manufacture	Government proving ground
Stowed equipment check	4.5.2.4	X	X
Fuel tank check	4.5.2.5	X	X
Vision devices and receptacle test	4.5.2.6	X	X
Turret control hydraulic fluid test	4.5.2.7	X	X
Boresight retention test, main sighting system	4.5.2.8	X	X
Boresight retention test, machine gun sighting system	4.5.2.9	X	X
Vehicle hydraulic system fluid test	4.5.2.10	X	X
Fire extinguisher control check	4.5.2.11	X	X
Gas particulate filter system	4.5.2.12	X	X
Comparison test	4.5.3		X
Cooling system test	4.5.3.2	**	X
Fuel system operational test	4.5.3.3		X
Ventilation test	4.5.3.4	**	X
Environmental tests	4.5.3.5		X
Speeds on grades check	4.5.3.6		X
Deep water fording test	4.5.3.7		X
Trench crossing test	4.5.3.8		X
Vertical obstacle test	4.5.3.9		X
Climbing test	4.5.3.10	**	X
Carbon dioxide distribution check	4.5.3.11		X

* In-process (see 4.3k) and Control test (see 4.5.2).
** Functional acceptance test.

4.5.1.3, delete second sentence and substitute:

"All hatches shall be checked for leakage."

PAGE 35

4.5.1.40.12, delete "hosizontal" and substitute "horizontal".

PAGE 37

4.5.1.41, second sentence delete "may" and substitute: "shall".

4.5.1.41, add the following to end of paragraph:

"Test performed on the indoor turret stand will be considered in-process inspection and shall be reverified as a control test using approved field test methods".

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4.5.1.42.6.3, add the following after subparagraph g.

"4.5.1.42.7 Gunner/commander night vision. To determine conformance to 3.6.14.6 place an aperture stop over the night vision channel of the gunners periscope. The stop shall limit the entrance of ambient light into the image intensifier tube to not more than 1/16 inch diameter. Upon actuation of the night vision gain switch, observation of a target image shall be possible. Rotating the night vision gain switch in a clockwise direction shall increase image intensity. Verify that the reticle is "off" when the reticle control knob is "off" and "on" when the control knob is "on". Reticle brightness shall increase with clockwise rotation of the control knob, and decrease with counterclockwise rotation. Repeat the procedure for the commanders periscope."

PAGE 42

4.5.1.43.5, after the last sentence in the paragraph add the following:

"4.5.1.44 Emergency mode firing. To determine conformance to 3.4.9.18 the blasting machine circuits shall be checked for sufficient power to fire the 165mm gun (see 3.4.9.16). The firing circuits from the blasting machine to the 165mm gun shall be interrupted (open) when the loaders safety switch is in the safe position and completed (closed) when in the fire position

4.5.1.45 Elevation shut-off valve. To determine conformance to 3.6.13.16 the gunners power switch shall be on, and without actuating the palm switch, the gunners control handle shall be tilted forward and backward to its stop. Gun movement in the elevation quadrant shall not be more than 2 mils.

4.5.1.46 Turret traverse brake. To determine conformance to 3.6.13.17 the turret shall be placed in a level position and the turret azimuth noted on the azimuth indicator. With the gunners power switch on, the gunners control handle shall be turned to its stop in a clockwise direction without actuating the palm switch. The azimuth indicator shall be observed for turret movement. The procedure shall be repeated in the counterclockwise direction. The turret shall then be power traversed no less than 10 degrees to the left with the gunners controls, stopped, the position observed on the azimuth indicator, and the gunners control handle again turned to its right and then left stops without depressing the palm switch. The procedure shall be repeated after power traverse to the right. The azimuth indicator shall show no movement of the turret during any of these checks.

4.5.1.47 Fuel shut-off valve. To determine conformance to 3.4.4.4 the fuel shut-off handle shall be actuated to the "off" position; the engine shall stop operating in not more than 30 seconds."

4.5.2, after the first sentence add the following:

"Tests annotated * in table VIII shall be re-verified as a control test."

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4.5.2.6, fourth sentence delete "receptalces" and substitute "receptacles".

PAGE 45

4.5.3.2, after the first sentence add the following:

"For acceptance test, operate the vehicle at ambient temperature and observe the engine and oil temperature indicators on the drivers panel. Verify that the temperature indicators remain in the green area during all stages of vehicle operation."

PAGE 46

4.5.3.4, subparagraph b, add the following:

"Conformance shall be determined during acceptance inspection by observing the vent blower operation when performing road tests and when the vehicle engine is at idle."

4.5.3.10, after the first sentence add the following:

"Conformance shall be determined during acceptance inspection by observing performance tests of 4.5.1.4, 4.5.1.33 and 4.5.1.35."

PAGE 47

5.1, delete "inaccordance" and substitute "in accordance".

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